

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I, Thanigavel (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thanigavel

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	OS.
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1.5	1.5	1.5	1.5	1.5
Analysis	20%	1	1	1	1	1
Code Quality	15%	1/2	1/2	1/2	1/2	1/2

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.

- Calculating the total number of usable host addresses after correction.
2. Two devices are configured with the following IPv6 addresses:
- Device A: 2001:db8::ff00:42:8329/64
 - Device B: 2001:db8:0:0:1::1/64
- Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
 - Determining the network prefix of each device.
 - Verifying whether both devices belong to the same subnet.
 - Identifying the addressing or prefix mismatch.
 - Optimizing the IPv6 configuration by assigning appropriate addresses.
 - Calculating the number of available host addresses within the corrected subnet.
3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
 - Calculating the number of usable and unused IP addresses.
 - Identifying inefficient subnet allocations.
 - Debugging the addressing plan by locating subnet allocation mistakes.
 - Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.
4. Two devices are configured with the following IPv6 addresses:
- Device A: 2001:db8::ff00:42:8329/64
 - Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.

- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. IPV4 Debugging:

-> Subnet -> 192.168.10.64 /26.

-> Network Address : 192.168.10.64.

-> Broadcast Address : 192.168.10.172.

-> Host Range : 192.168.10.65 - 126.

-> Error : IP and Gateway in different subnets.

-> Correct IP : 192.168.10.2 /26.

-> Gateway : 192.168.10.1.

-> Usable Hosts : 62.

2. IPV6 Debugging:

-> Expanded A : ~~2001:0b08:0000:0000:0000:ff00:0000:8329~~

-> Expanded B : 2001:0b08:0000:0000:0001:0000:0000:0001

-> Network prefix : 2001:db8:0:0:104

-> Same subnet : Yes.

-> Correct Addresses : ::10 and ::20.

-> Available Hosts : 2^{64} .

3) Subnet Analysis:

Subnet	Network Address	Host Range	Broadcast	Usable hosts
126	192.168.1.0	192.168.1.1 - 192.168.1.62	192.168.1.63	62
127	192.168.1.64	65-94	192.168.1.95	30
128	192.168.1.96	97-110	192.168.1.111	14

issue: Unequal IP allocation.

Solution: use VLSM better utilization.

4. IPV6 Debugging.

Given:

Device A: 2001:db8:ff00:8329 /64.

Device B: 2001:db8:0000:1:1 /64.

Ans: (i) Expanded addresses:

A: 2001:0db8:0000:0000:0000:ff00:0092:8329

B: 2001:0db8:0000:0000:0001:0000:0000:0001

(2) Network profile:-

Both: $2001:db8:0:0::/64$.

③ Same subnet:

*yes.

4) Error:-

-> No addressing error: check interface routing or firewall.

5) Correct configuration:

-> A: $2001:db8:0:0::/64$.

-> B: $2001:db8:0:0::/20$ /64.

6) Available hosts:

-> $2^{64} = 18,446,744,073,709,551,616$.

④ Routing table.

-> Matching Route: - $192.168.2.0/24$.

-> Next Hop: - $10.0.0.3$.

-> Destination Valid: yes.

-> If No match: Use Default Route ($10.0.0.1$).